Pre-Calculus

functions

A **function** takes an input and returns exactly one output. Write $f(x)=2x+1$ and read it as "f of x".

$f(x)$ is not f times x. It names the output that f produces from the input x.

To **evaluate**, replace every x with the input: $f(3)=2(3)+1=7$.

domain	every input the function accepts
range	every output it can produce

Finding a domain — three blockers

Assume every real number works, then remove the values that break one of these:

1. **division by zero** ⇒ set the denominator to zero and exclude those values. $\frac{1}{x-3}$ excludes $x=3$.
2. **even root of a negative** ⇒ the inside must be ≥ 0. $\sqrt{x-2}$ needs $x \geq 2$.
3. **logarithm of a non-positive** ⇒ the inside must be >0.

No blocker present means the domain is all real numbers. Polynomials always have that domain.

Even and odd

even	$f(-x)=f(x)$	mirror across the y-axis
odd	$f(-x)=-f(x)$	same after a 180° turn
neither	neither holds	most functions

THE TEST — SUBSTITUTE $-x$ AND COMPARE

$f(x)=x^2-4$: $f(-x)=(-x)^2-4=x^2-4$. That equals $f(x)$, so even.

$f(x)=x^3-x$: $f(-x)=-x^3+x=-(x^3-x)$. That equals $-f(x)$, so odd.

Compare against both $f(x)$ and $-f(x)$ before concluding "neither".

Transformations

Start from $y=f(x)$. Each change moves or reshapes the graph.

$f(x)+k$	up k *(down if $k<0$)*
$f(x-h)$	right h note the sign flip
$-f(x)$	flip across the x-axis
$f(-x)$	flip across the y-axis
$a\,f(x)$	stretch vertically by a

Changes inside the brackets do the opposite of what they look like. $f(x-3)$ moves the graph right 3, not left. Inside affects the input, so it works backwards.

Composition

$(f \circ g)(x)$ means $f(g(x))$: run g first, then feed its output into f.

WORKED: $f(x)=2x+1$, $g(x)=x^2$

$(f \circ g)(3)$: $g(3)=9$, then $f(9)=2(9)+1=19$

$(g \circ f)(3)$: $f(3)=7$, then $g(7)=7^2=49$

Order matters. $19 \neq 49$, so $f \circ g$ and $g \circ f$ are different functions.

Inverses

The **inverse** f^{-1} undoes f. If f turns 3 into 7, then f^{-1} turns 7 into 3.

THE PROCESS

1. **Replace** $f(x)$ with y.
2. **Swap** x and y.
3. **Solve** for y.
4. **Rename** it $f^{-1}(x)$.

WORKED: INVERT $f(x)=2x+1$

$y=2x+1 \;\rightarrow\;$ swap: $x=2y+1 \;\rightarrow\;$ solve: $y=\frac{x-1}{2}$

$$f^{-1}(x)=\frac{x-1}{2}$$

check: $f(3)=7$, and $f^{-1}(7)=\frac{6}{2}=3$ ✓

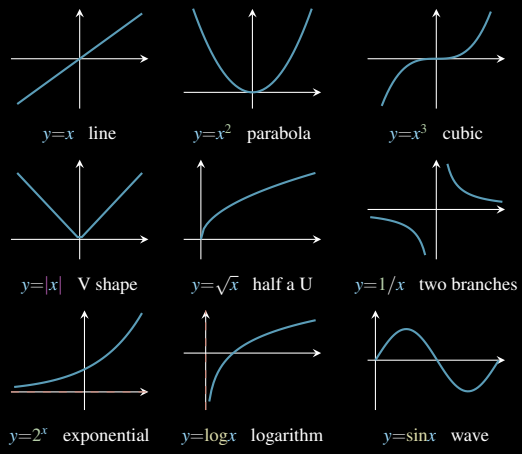
f^{-1} does not mean $\frac{1}{f}$. The -1 here names the inverse function, not a negative exponent.

Piecewise functions

A **piecewise function** uses different rules on different parts of the domain. Pick the rule whose condition your input satisfies, then use only that rule.

the parent graphs

Every graph you meet is one of a small set of **parent functions**, moved or stretched. Learn these nine shapes and the transformations above will let you sketch almost anything.



parent	domain	range	what to remember
$y=x$	all reals	all reals	straight through the origin
$y=x^2$	all reals	$y \geq 0$	a U, even, never negative
$y=x^3$	all reals	all reals	an S, odd, flattens at the origin
$y=\lvert x\rvert$	all reals	$y \geq 0$	a V with a sharp corner
$y=\sqrt{x}$	$x \geq 0$	$y \geq 0$	stops dead at the origin
$y=\frac{1}{x}$	$x \neq 0$	$y \neq 0$	both axes are asymptotes
$y=2^x$	all reals	$y>0$	never reaches $y=0$
$y=\log x$	$x>0$	all reals	never reaches $x=0$
$y=\sin x$	all reals	$-1 \leq y \leq 1$	repeats every 2π

The dashed red lines are asymptotes — the curve approaches them forever without arriving.

2^x and $\log x$ are mirror images across the line $y=x$. They are inverse functions, which is exactly why their domains and ranges swap over.

polynomial and rational behaviour

END BEHAVIOUR — THE LEADING TERM DECIDES

Only the highest-power term matters far from the origin. Read its degree and sign.

even degree — both ends point the same way. Positive lead: both up.

odd degree — the ends point opposite ways. Positive lead: down on the left, up on the right.

Asymptotes — three kinds

An **asymptote** is a line the graph approaches but never reaches.

V **vertical** ⇒ where the denominator is zero after cancelling. A factor that cancels gives a **hole**, not an asymptote.

H **horizontal** ⇒ compare degrees. Top smaller: $y=0$. Equal: the ratio of leading coefficients. Top larger: none.

S **slant** ⇒ when the top degree is exactly one more than the bottom. Divide to find the line.

WORKED: $\dfrac{2x^2+3x-1}{x^2-4}$

vertical: $x^2-4=0$ gives $x=2$ and $x=-2$, and neither cancels

horizontal: degrees are equal, both 2, so take the leading coefficients:

$$y=\frac{2}{1}=2$$

slant: none, because the degrees are equal rather than differing by one

A graph may cross a horizontal asymptote. The rule is about behaviour far out, not a barrier.

finding the zeros

Four words for one thing. A **zero**, a **root**, an **x-intercept**, and a **solution of $f(x)=0$** all name the same number: an input where the output is zero, so the graph touches the x-axis.

THE PROCESS

1. **Set the function equal to zero.**
2. **Solve** by whichever method fits the type.
3. **Discard** any answer outside the domain.
4. **Check** each survivor in the original function.

Which method? — read the function type

A **linear** ⇒ solve directly. Exactly one zero, unless the line is horizontal.

B **quadratic** ⇒ factor, or use the formula. The discriminant says how many before you start.

C **polynomial, degree ≥ 3** ⇒ rational root theorem, then divide and factor what is left.

D **rational** ⇒ set the numerator to zero, then discard anything that also zeroes the denominator.

E **exponential** ⇒ no zeros ever. b^x is always positive, which is why $y=0$ is its asymptote.

F **logarithm** ⇒ set the inside equal to 1, because $\log 1=0$.

G **trigonometric** ⇒ infinitely many, spaced one period apart. Give the ones in the interval asked for.

ONE EXAMPLE OF EACH

A $3x-12=0 \;\Rightarrow\; x=4$
B $x^2-6x+5=0 \;\Rightarrow\; (x-1)(x-5)=0 \;\Rightarrow\; x=1,\,5$
C $x^3-4x=0 \;\Rightarrow\; x(x-2)(x+2)=0 \;\Rightarrow\; x=0,\,2,\,-2$
F $\log(x-3)=0 \;\Rightarrow\; x-3=1 \;\Rightarrow\; x=4$
G $\sin x=0$ on $[0,2\pi) \;\Rightarrow\; x=0,\,\pi$

CASE D IS THE TRAP — $f(x)=\dfrac{x^2-4}{x-2}$

numerator zero at $x=2$ and $x=-2$. But $x=2$ also makes the denominator zero.

$x=2$ is a **hole**, not a zero — the function has no value there at all.

the only zero is $x=-2$

Always check a candidate against the denominator before accepting it.

Always factor first. A factored function shows its zeros directly: each factor set to zero gives one, by the zero product rule.

the key points of a graph

To sketch a curve, or to describe one, you need a small fixed list of features. Find them in this order.

THE PROCESS

1. **y-intercept** — set $x=0$. There is at most one, because a function gives one output per input.
2. **x-intercepts** — set $y=0$ and find the zeros. There may be many.
3. **Turning points** — where the curve changes direction.
4. **Asymptotes** — lines the curve approaches but never reaches.
5. **End behaviour** — what happens far left and far right.

A **turning point** is where a graph stops rising and starts falling, or the reverse. It is the peak of a hill or the bottom of a valley. Two you can find exactly, without calculus:

parabola vertex $x=-\dfrac{b}{2a}$, then substitute to get y

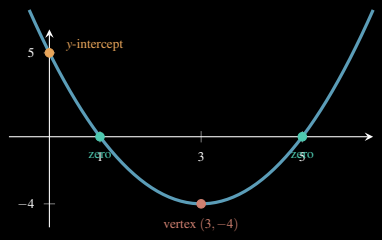
by symmetry the vertex sits halfway between the two zeros

A polynomial of degree n has at most $n-1$ turning points. A parabola has one, a cubic has at most two. That is a useful check on a sketch.

For any other curve, locating turning points exactly needs calculus. That is one of the main things calculus is for, and it is where this material leads next.

WORKED: FULLY DESCRIBE $y=x^2-6x+5$

1. y-intercept: $y=0-0+5=5$, so the point $(0,5)$
2. x-intercepts: $(x-1)(x-5)=0$, so $x=1$ and $x=5$
3. vertex: $x=-\dfrac{-6}{2(1)}=3$, and $y=9-18+5=-4$, so $(3,-4)$
(check by symmetry: halfway between 1 and 5 is 3 ✓)
4. asymptotes: none — polynomials have no asymptotes
5. end behaviour: degree 2 is even and the lead is positive, so both ends rise
That is enough to sketch it: a U crossing at 1 and 5, bottoming at $(3,-4)$, passing through $(0,5)$.



the five features, on one curve: two zeros, the vertex halfway between them, and the y-intercept

sketching a polynomial

A parabola has one turn. Higher degrees have more. You can locate all of them without calculus, using three facts you already have: the zeros, the end behaviour, and the sign in between.

THE PROCESS

1. **Factor** and list the zeros. Those are the only places the curve can cross.
2. **Check each zero's multiplicity** — does the curve cross there, or touch and turn back?
3. **Find the y-intercept** by setting $x=0$.
4. **Read the end behaviour** from the leading term.
5. **Test one point in each interval** between the zeros to get the sign.
6. **Join the dots.** A turning point sits between each pair of neighbouring zeros.

Multiplicity — cross or touch?

The **multiplicity** of a zero is how many times its factor repeats.

factor	multiplicity	at that zero the curve
$(x-a)$	1, odd	crosses straight through
$(x-a)^2$	2, even	touches and turns back
$(x-a)^3$	3, odd	crosses, but flattens on the way

Odd multiplicity crosses, even multiplicity touches. An even power can never change sign, so the curve cannot get to the other side.

A touch point is always a turning point. If the curve comes down to the axis and does not cross, it must turn around there.

WORKED: $y=(x+1)^2(x-2)$

zeros: $x=-1$ with multiplicity 2, $x=2$ with multiplicity 1

so the curve touches at $x=-1$ and crosses at $x=2$

check by testing either side of -1: $f(-1.2)\approx -0.13$ and $f(-0.8)\approx -0.11$

Both negative, so the curve never got above the axis — it touched and came back ✓

Worked: a cubic with two turns

Sketch $y=(x+2)(x-1)(x-3)$, which expands to $y=x^3-2x^2-5x+6$.

1–3. ZEROS, MULTIPLICITY, INTERCEPT

zeros: $x=-2, 1, 3$ — each factor appears once, so the curve crosses at all three

y-intercept: $y=(2)(-1)(-3)=6$

↓

4. END BEHAVIOUR

degree 3 is odd and the leading coefficient is positive, so the ends point opposite ways: down on the left, up on the right.

↓

5. SIGN IN EACH INTERVAL — TEST ONE POINT

three zeros cut the line into four intervals. Test any point in each:

$x<-2$	$x=-3$	$y=-24$	below the axis
$-2<x<1$	$x=0$	$y=6$	above
$1<x<3$	$x=2$	$y=-4$	below
$x>3$	$x=4$	$y=18$	above

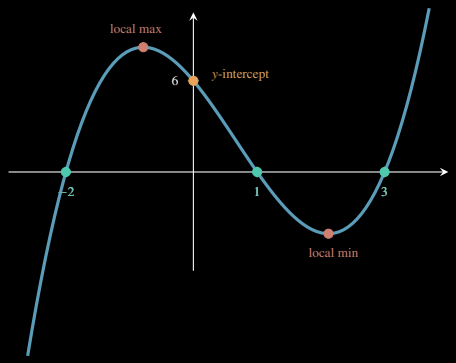
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6. TURNING POINTS — BETWEEN NEIGHBOURING ZEROS

The curve rises from below to above between -2 and 1, so it must turn back down somewhere in there. That is a local maximum, near $x\approx -0.8$, at about $y\approx 8.2$.

It falls from above to below between 1 and 3, so it turns back up: a local minimum near $x\approx 2.1$, at about $y\approx -4.1$.

These positions are approximate. A degree-3 curve has at most $3-1=2$ turns, and you know they exist and roughly where — but pinning them down exactly needs calculus.



three zeros, the y-intercept, and the two turning points — one between each neighbouring pair of zeros

maximums and minimums

A **maximum** is a peak, a **minimum** is a valley. The only thing that ever confuses people here is local against absolute.

local max	higher than everything nearby. A hilltop, even if taller hills exist elsewhere.
local min	lower than everything nearby
absolute max	the highest point anywhere on the whole graph or interval
absolute min	the lowest point anywhere

Every absolute extreme is also a local one, but not the other way round. In the cubic above, $y\approx 8.2$ is a local max, yet the curve climbs past it forever on the right — so there is no absolute maximum.

Finding them without calculus

A **parabola** ⇒ exact, from the vertex: $x=-\frac{b}{2a}$. Opens up means a minimum, opens down means a maximum.

B **a zero of even multiplicity** ⇒ exact, because a touch point is a turning point and its value is 0.

C **any other curve** ⇒ approximate. You know one turn sits between each neighbouring pair of zeros. Read it off a sketch or a calculator.

Exact turning points for cubics and beyond need calculus. Say "about 8.2 near $x\approx -0.8$" rather than inventing an exact value.

ON A CLOSED INTERVAL, CHECK THE ENDS

Asked for the extremes on an interval such as $[0,4]$, collect three kinds of candidate:

1. every turning point inside the interval **2.** the left endpoint **3.** the right endpoint

Then compare the values and take the largest and smallest.

The endpoints are the ones people forget, and they very often win. On $[0,4]$ the cubic above gives $f(0)=6$, $f(4)=18$, and the interior minimum ≈ -4.1. The absolute maximum is 18, at the right endpoint — not at any turn.

A parabola's vertex is the one case you can always state exactly, which is why so many word problems about greatest area or least cost end in a quadratic.

polynomial and rational inequalities

The same sign-testing that sketches a curve also solves an inequality. You are asking where the curve sits above or below the axis.

THE PROCESS

1. Move everything to one side, so the other side is zero.
2. Factor and find the zeros. For a fraction, also find where the denominator is zero.
3. Mark all of those on a number line. They split it into intervals.
4. Test one point in each interval.
5. Keep the intervals whose sign matches what was asked.

Never multiply an inequality by something containing the variable. You do not know its sign, so you do not know whether to flip. Sign-testing avoids the problem entirely.

WORKED: $x^2-5x+6>0$

factor: $(x-2)(x-3)>0$, so the zeros are 2 and 3

test each of the three intervals:

$x<2$	$x=0$	$+6$	keep
$2<x<3$	$x=2.5$	-0.25	reject
$x>3$	$x=4$	$+2$	keep

$$(-\infty, 2)\cup(3, \infty)$$

strict $>$ means the zeros themselves are excluded, so round brackets.

For a rational inequality, mark the denominator's zeros too — the sign can flip there as well, even though the curve never touches the axis. Those points are always excluded, whatever the inequality sign says.

conic sections

Four curves come from slicing a cone. Identify one by looking at the squared terms.

curve	equation	the tell
circle	$(x-h)^2+(y-k)^2=r^2$	both squared, added, same coefficient
parabola	$y=a(x-h)^2+k$	only one variable squared
ellipse	$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$	both squared, added, different coefficients
hyperbola	$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$	both squared, subtracted

In every form (h,k) is the **centre**, or the **vertex** for a parabola. The signs inside the brackets flip: $(x-3)^2$ means the centre sits at $x=+3$.

If the equation is not in one of these forms, complete the square on each variable to put it there.

right-triangle trigonometry

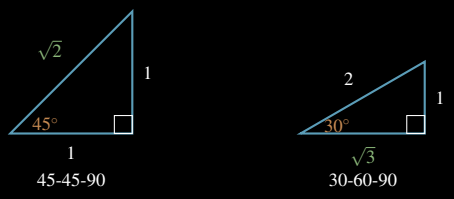
In a right triangle, three ratios relate an angle to the sides. Remember them as SOH CAH TOA.

$$\sin\theta=\frac{\text{opposite}}{\text{hypotenuse}} \qquad \cos\theta=\frac{\text{adjacent}}{\text{hypotenuse}} \qquad \tan\theta=\frac{\text{opposite}}{\text{adjacent}}$$

"Opposite" and "adjacent" depend on which angle you picked. The hypotenuse never changes, but the other two swap when you switch angles.

The two special triangles

These give exact answers without a calculator, so they appear constantly.



45-45-90 30-60-90

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$
$30°=\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
$45°=\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$60°=\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$

Notice the symmetry: $\sin30°=\cos60°$. The two angles add to $90°$, and the opposite side of one is the adjacent side of the other.

the unit circle

The **unit circle** has radius 1 and centre at the origin. It extends trigonometry past $90°$.

$$\cos\theta=x \qquad \sin\theta=y$$

Every point on the circle is $(\cos\theta,\sin\theta)$. Angles are measured counterclockwise from the positive x-axis.

Radians measure angles by arc length instead of degrees. A full circle is 2π radians, so

$$180°=\pi \text{ radians} \qquad \text{degrees} \to \text{radians}: \text{ multiply by } \frac{\pi}{180}$$

quadrant	positive there	because
I	all	$x>0$, $y>0$
II	sin only	$y>0$ but $x<0$
III	tan only	both negative, so the ratio is positive
IV	cos only	$x>0$ but $y<0$

Derive the signs, do not memorize them. Ask whether x and y are positive in that quadrant, then read off which ratio survives.

trigonometric identities and equations

THE ONE IDENTITY EVERYTHING ELSE COMES FROM

$$\sin^2\theta+\cos^2\theta=1$$

This is the Pythagorean theorem on the unit circle: the legs are $x=\cos\theta$ and $y=\sin\theta$, and the hypotenuse is the radius 1.

quotient $\tan\theta=\frac{\sin\theta}{\cos\theta}$

reciprocals $\csc=\frac{1}{\sin}$, $\sec=\frac{1}{\cos}$, $\cot=\frac{1}{\tan}$

double angle $\sin2\theta=2\sin\theta\cos\theta$

$\cos2\theta=\cos^2\theta-\sin^2\theta$

THE PROCESS — SOLVE A TRIG EQUATION

1. **Isolate** the trig function.
2. **Find** one angle that works, using a special triangle.
3. **Find every other angle** in the interval with the same value.
4. **Check** each against the interval you were given.

WORKED: SOLVE $\sin x=\frac{1}{2}$ ON $[0,2\pi)$

sin is the y-coordinate, and $y=\frac{1}{2}$ at two places on the circle.

first angle: $x=\frac{\pi}{6}$ *(from the 30-60-90 triangle)*

y is also positive in quadrant II, and the matching angle there is $\pi-\frac{\pi}{6}=\frac{5\pi}{6}$

$$x=\frac{\pi}{6} \quad \text{or} \quad x=\frac{5\pi}{6}$$

A trig equation almost never has just one answer. Sweep the whole interval before stopping.

graphs of the trigonometric functions

sin and cos repeat forever. Every transformation you need is visible in one general form:

$$y=A\sin\big(B(x-C)\big)+D$$

letter	name	what it does
$\lvert A\rvert$	**amplitude**	height from the middle to a peak
$\frac{2\pi}{B}$	**period**	how far before the pattern repeats
C	**phase shift**	slide left or right
D	**vertical shift**	the new middle line

B is not the period. The period is $\frac{2\pi}{B}$. A larger B squeezes the wave, making the period shorter.

Factor before reading C. In $\sin(2x-\pi)$ the shift is not π. Factor the B out first.

WORKED: $y=3\sin(2x-\pi)+1$

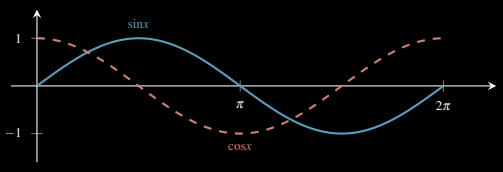
1. factor the inside: $2x-\pi=2\left(x-\dfrac{\pi}{2}\right)$

2. read the four numbers: $A=3$, $B=2$, $C=\dfrac{\pi}{2}$, $D=1$

3. translate them:

amplitude 3 period $\dfrac{2\pi}{2}=\pi$ shift $\dfrac{\pi}{2}$ right middle line $y=1$

So the wave runs from $y=-2$ up to $y=4$, completing a full cycle every π.



both have amplitude 1 and period 2π. cos is sin shifted left by $\frac{\pi}{2}$ — the same wave, started earlier

tan behaves differently: its period is π, not 2π, and it has a vertical asymptote wherever $\cos x=0$.

inverse trigonometric functions

$\sin x=\frac{1}{2}$ has infinitely many answers, so sin has no inverse until you restrict its domain. Each inverse therefore returns one angle from a fixed range.

function	returns an angle in	quadrants
$\arcsin x$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	IV and I
$\arccos x$	$[0, \pi]$	I and II
$\arctan x$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	IV and I

This is why a calculator can give the "wrong" angle. It returns the one angle in that range. If your problem sits in another quadrant, adjust it yourself — the same trap as converting to polar coordinates.

arcsin undoes sin only inside the restricted range. $\arcsin\left(\sin\frac{5\pi}{6}\right)$ is $\frac{\pi}{6}$, not $\frac{5\pi}{6}$, because $\frac{5\pi}{6}$ is outside $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

triangles that are not right-angled

SOH CAH TOA needs a right angle. Two laws handle every other triangle.

Which law? — count what you are given

A **Law of Sines** ⇒ you have a matching angle and opposite side. Cases **AAS**, **ASA**, **SSA**.

B **Law of Cosines** ⇒ you do not. Cases **SAS** and **SSS**.

The test is one question: can you see an angle and the side directly across from it, both known? If yes, use Sines. If no, use Cosines.

Law of Sines $\quad \dfrac{a}{\sin A}=\dfrac{b}{\sin B}=\dfrac{c}{\sin C}$

Law of Cosines $\quad c^2=a^2+b^2-2ab\cos C$

Each side pairs with the angle opposite it. Lower-case a sits across from upper-case A.

The Law of Cosines is the Pythagorean theorem with a correction term. When $C=90°$, $\cos C=0$, the last term vanishes, and it becomes $c^2=a^2+b^2$ exactly.

WORKED (A): $A=30°$, $B=45°$, $a=10$

a and A are a matching pair, so use Sines:

$$\frac{10}{\sin 30°}=\frac{b}{\sin 45°} \implies b=\frac{10\sin 45°}{\sin 30°}=\frac{10\left(\frac{\sqrt{2}}{2}\right)}{\frac{1}{2}}$$

$$b=10\sqrt{2}\approx 14.14$$

Sanity check: B is the larger angle, so b must be the longer side. $14.14>10$ ✓

↓

WORKED (B): $a=5$, $b=7$, $C=60°$

The known angle sits between the two known sides, so no matching pair exists. Use Cosines:

$$c^2=5^2+7^2-2(5)(7)\cos 60°=25+49-70\left(\tfrac{1}{2}\right)=74-35=39$$

$$c=\sqrt{39}\approx 6.24$$

Sanity check: c must sit between $7-5=2$ and $7+5=12$. It does ✓

SSA can give two triangles, one, or none. Two different angles share the same sine, so after finding one, check whether its supplement also fits inside $180°$. This is called the **ambiguous case**.

Sum and difference identities

These build exact values for angles the special triangles do not cover.

$$\sin(A\pm B)=\sin A\cos B\pm\cos A\sin B$$

$$\cos(A\pm B)=\cos A\cos B\mp\sin A\sin B$$

The cos signs are reversed. A plus inside gives a minus in the middle. This is the single most-missed detail in the section.

WORKED: EXACT VALUE OF $\sin 75°$

$75°$ is not a special angle, but $75=45+30$ and both of those are:

$$\sin 75°=\sin 45°\cos 30°+\cos 45°\sin 30°$$

$$=\left(\tfrac{\sqrt{2}}{2}\right)\left(\tfrac{\sqrt{3}}{2}\right)+\left(\tfrac{\sqrt{2}}{2}\right)\left(\tfrac{1}{2}\right)=\frac{\sqrt{6}}{4}+\frac{\sqrt{2}}{4}=\frac{\sqrt{6}+\sqrt{2}}{4}$$

check: this is about 0.966, and $\sin 75°$ should be just under 1 ✓

Setting $A=B$ turns these into the double-angle identities above. They are one family, not two.

combining functions

Two functions can be added, subtracted, multiplied or divided, exactly like numbers.

$(f+g)(x)$	$f(x)+g(x)$
$(f-g)(x)$	$f(x)-g(x)$
$(fg)(x)$	$f(x)\cdot g(x)$
$\left(\dfrac{f}{g}\right)(x)$	$\dfrac{f(x)}{g(x)}$, and $g(x)\neq 0$

The new domain is what both originals allow, minus anything the operation itself forbids. Division adds the rule that $g(x)$ may not be zero.

WORKED: $f(x)=2x+1$, $g(x)=x^2$

$(f+g)(x)=2x+1+x^2$ so $(f+g)(3)=6+1+9=16$

$(fg)(x)=(2x+1)x^2=2x^3+x^2$ so $(fg)(3)=54+9=63$

This is not composition. $(fg)(3)$ multiplies the two outputs and gives 63. $(f\circ g)(3)$ feeds one into the other and gives 19. Different operations, similar notation.

average rate of change

Slope measures how fast a line climbs. For a curve, the steepness changes, so you measure it between two points instead.

$$\text{average rate of change} = \frac{f(b)-f(a)}{b-a}$$

This is exactly the slope formula. It is the slope of the straight line joining the two points on the curve, called the **secant line**.

WORKED: $f(x)=x^2$ **ON** $[1,3]$

$f(1)=1$ and $f(3)=9$:

$$\frac{9-1}{3-1}=\frac{8}{2}=4$$

So across that stretch the curve rises 4 units of y for each unit of x, on average.

It is not the steepness at any single point. At $x=1$ the curve is much flatter than 4, and at $x=3$ much steeper. The answer is the average of everything between.

Shrink the interval until the two points nearly touch, and the average rate becomes the steepness at a single point. That limit is the derivative, and it is the first idea in calculus. Everything in this document has been leading here.

polar coordinates

Polar coordinates locate a point by distance and direction instead of across and up. Write (r,θ).

polar $\to$ **rectangular** $x=r\cos\theta, \quad y=r\sin\theta$

rectangular $\to$ **polar** $r=\sqrt{x^2+y^2}, \quad \tan\theta=\frac{y}{x}$

WORKED: $(3,4)$ **TO POLAR**

$r=\sqrt{3^2+4^2}=\sqrt{25}=5$

$\tan\theta=\frac{4}{3}$, so $\theta=\arctan\frac{4}{3}\approx 53.13°$

Check the quadrant. arctan only reports angles in quadrants I and IV. If $x<0$, add $180°$.

vectors

A **vector** has both size and direction. Write it in components as $\langle a,b\rangle$.

magnitude	$\|\langle a,b\rangle\|=\sqrt{a^2+b^2}$
add	add matching components
scalar multiply	multiply every component
unit vector	divide the vector by its own magnitude

WORKED: $\vec{v}=\langle 3,4\rangle$

magnitude: $\|\vec{v}\|=\sqrt{9+16}=\sqrt{25}=5$

unit vector: divide each component by 5:

$$\hat{v}=\left\langle \frac{3}{5}, \frac{4}{5}\right\rangle$$

check: $\sqrt{\left(\frac{3}{5}\right)^2+\left(\frac{4}{5}\right)^2}=\sqrt{\frac{9}{25}+\frac{16}{25}}=\sqrt{1}=1$ ✓

A unit vector keeps the direction and throws away the size. That is why dividing by the magnitude always lands on 1.

Legend

Colours: what we tell you
orange — a new word
teal — the answer
red — a common mistake
white — explanation
boxes stack top to bottom
↓ means "next step"

Colours: inside the maths
blue — a letter x,y
green — a number $3,7$
yellow — a function f,g
pink — brackets $(\,)$
grey — signs $+\,-\,=$

Comparing
$=$ is equal to
$\neq$ is not equal to
$<$ is less than
$>$ is greater than
$\leq$ less than or equal
$\geq$ greater than or equal
$\approx$ is about

Operations
$\cdot$ times
$\pm$ plus or minus
$\sqrt{\ }$ square root
$\sqrt[n]{\ }$ nth root
$|x|$ absolute value
x^{-1} one over x
f^{-1} inverse function

Names
π about 3.14159
θ an angle
i $\sqrt{-1}$
m slope
a_n the nth term
$\Rightarrow$ therefore